

AI201 Mini Project Report

Anticipating Local Typhoon Impacts: Track Clustering, Rapid Intensification, and the Bantay-Bagyo Index (BBI)

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1. INTRODUCTION

On average, the Philippine Area of Responsibility experiences about 20 tropical cyclones in a year [1]. Recent typhoons such as Kristine (Trami, 2024) and Uwan (Fung-wong, 2025) have caused severe damage across the Philippines. Kristine brought torrential rains that displaced over 630,000 people and killed at least 162, with damages reaching P14 billion [2]. Typhoon Uwan, a Category-4 storm, affected more than 7 million people nationwide, damaged over 274,000 houses, and caused at least 33 deaths despite large-scale evacuations [3]. These events highlighted the risks of rapid intensification and the need for better local preparedness tools, as even well-tracked storms can cause widespread destruction if their local impacts are not clearly forecast [2][3]. They underscore the urgent need to translate storm forecasts into actionable, community-level impact predictions.

Local governments in the Philippines face critical challenges in typhoon preparedness due to both meteorological surprises and gaps in forecast interpretation. One major issue is the rapid intensification of storms just before landfall, which shortens the window for evacuations and local response planning [2]. Super Typhoon Uwan (Fung-wong) in 2025 rapidly intensified into a Category-4 storm before landfall in Aurora, bringing widespread destruction to 16 regions and affecting over 7 million people nationwide [3]. Another challenge arises when storms deviate from predicted tracks, affecting regions that may not have prepared adequately. For instance, Typhoon Kristine (Trami) in 2024 triggered deadly floods and landslides in Batangas and other parts of Luzon after a slower-than-expected landfall, catching communities off guard [2]. These forecast surprises are frequently acknowledged in post-disaster reports and media assessments [2].

While agencies like PAGASA provide forecasts on track and intensity, these are not directly translated into actionable, localized impact projections for provincial and municipal governments. As one official noted, “good forecasts don’t always translate to good response,” since technical bulletins are often difficult for communities and decision-makers to interpret in terms of real-world consequences [4]. This reveals a significant gap between storm-scale forecasts and the ability to anticipate and communicate LGU-level impacts. In response, PAGASA and partner institutions are working to implement impact-based forecasting systems that aim to link meteorological data with estimates of expected damage

and human consequences [4]. This shift is especially important in light of recent events like Typhoons Kristine (2024) and Uwan (2025), where accurate hazard forecasts did not fully convey the extent of on-the-ground devastation [2][3].

This study aims to address these challenges by integrating storm behavior analysis with impact prediction tailored for local decision-making. The project combines track clustering, rapid intensification detection, and a locally informed impact metric to better anticipate the effects of typhoons. As a first step, Typhoon Track Clustering and Rapid Intensification (RI) Prediction were performed. Track clustering is important to identify dominant path patterns of tropical cyclones. This helps disaster-risk reduction initiatives identify regions repeatedly exposed to storms and forecasters use probability to predict which areas are likely to be affected by new tropical cyclones. Rapid intensification is defined as a sudden increase in tropical cyclone intensity, typically measured as an increase of greater than or equal to 30 knots in maximum sustained winds over 24 hours. RI poses a significant forecasting challenge, as it can greatly increase the destructive potential of a storm in a short time. Accurate identification of RI events is crucial for early warning and disaster preparedness.

At the core of this effort is the Bantay-Bagyo Index (BBI), a composite proxy that estimates a storm’s potential danger to specific communities by combining storm characteristics (e.g., peak wind speed, pressure, storm speed, size, rapid intensification history), impact reports (e.g., population affected, evacuations, casualties, power outages, flooding, state of calamity), and local vulnerability indicators (e.g., poverty indicators and city-level hazard area proportions). By linking meteorological features with human impact patterns, the project supports the development of more actionable, LGU-level preparedness insights.

1.1 Objectives

This study aims to enhance local typhoon preparedness by integrating storm behavior analysis with impact forecasting. The specific objectives are to:

1. Cluster typhoon tracks to identify recurring storm trajectories or archetypes using unsupervised learning techniques.
2. Model rapid intensification (RI) events using machine learning to classify periods when storms are likely to

intensify significantly within 24 hours.

3. Develop the Bantay-Bagyo Index (BBI) as a composite LGU–storm impact proxy that integrates storm features, observed impacts, and local vulnerability indicators.
4. Apply and evaluate the framework using Metro Manila as a case study, while ensuring that the methods and insights are scalable and adaptable for nationwide implementation.

1.2 Scope and Limitation

Track Clustering focuses on tropical cyclone tracks in the Philippines using IBTrACS data from 2000–2025, based on time and geographic coordinates of typhoon centers to identify dominant track patterns using clustering. Limitations include having the spatial resolution limited by IBTrACS coordinate precision and only storm center positions and times are used for clustering.

Rapid Intensification also focuses on tropical cyclones in the Philippines using IBTrACS data from 2000–2025. Using features such as pressure, wind radii, eye of the storm, gust, storm speed, and storm direction to predict rapid intensification. The maximum sustained winds were used to identify rapid intensification over 24-hour intervals. The missing values and exclusion of environmental variables not available in IBTrACS may affect the prediction.

The Bantay-Bagyo Index (BBI) in this study is developed and tested using data specific to Metro Manila, a region selected for its relative data richness and detailed disaster reporting. The BBI integrates storm characteristics from typhoons that affected Metro Manila between 2020 and 2025, alongside local vulnerability indicators sourced from publicly available PSA datasets and hazard exposure data from Project NOAH. Impact data are derived from NDRRMC disaster reports within the same period. While the current application focuses on Metro Manila as a case study, the design of the BBI is intended to be scalable for nationwide use, potentially down to the barangay level, depending on data availability and granularity in other regions.

1.3 Contributions of the Study

This study advances localized, impact-based typhoon forecasting by integrating typhoon track clustering, rapid intensification (RI) modeling, and the development of the Bantay-Bagyo Index (BBI) — a composite metric for estimating storm impact severity at the local level. By linking storm-scale behavior to community-scale outcomes, the study offers a practical tool for supporting disaster preparedness, evacuation planning, and emergency response, particularly among local government units (LGUs). The BBI helps LGUs anticipate impacts more accurately and can serve as the basis for a live decision-support dashboard during storm events. Track clustering enables identification of recurring storm pathways, while RI modeling supports early warnings for storms that may intensify abruptly near landfall — both addressing known gaps in existing forecast systems. The system also has potential applications in insurance risk estimation, disaster financing, and public communication by translating technical forecasts into concrete, localized impacts.

Beyond operational use, the framework contributes to disaster risk reduction policy, climate adaptation planning, and urban resilience efforts. It is designed to be scalable — applicable not only to highly urbanized regions like Metro Manila, but also to barangay-level forecasting where data is available. The methodology also adds value to academic research on tropical cyclone dynamics and serves humanitarian actors seeking to optimize impact-based early warning systems.

2. RELATED WORK

Understanding typhoon behavior and its impacts has been the focus of extensive research, ranging from the clustering of tropical cyclone tracks to the prediction of rapid intensification and assessment of human vulnerability. Prior studies have employed a variety of statistical and machine learning approaches, including K-means and fuzzy c-means clustering [5, 6, 7, 8, 20, 21, 22, 23], ensemble and probabilistic models for rapid intensification [9, 10, 24, 11, 25], and vulnerability-based risk modeling [12, 13, 14]. These investigations have advanced our understanding of storm trajectories, intensity evolution, rainfall prediction, and the socio-environmental factors that shape cyclone impacts. However, many of these approaches focus on hazard characterization, rainfall estimation, or intensity prediction without fully integrating localized exposure and vulnerability. The current study builds on these insights by combining track clustering, rapid intensification prediction, rainfall modeling, and human impact assessment into a unified framework. A summary of the related work, including methods, key findings, and the gaps addressed by the Bantay-Bagyo Index (BBI), is presented in Table 1.

3. METHODOLOGY

3.1 Dataset and Preprocessing

Historical tropical cyclone track data from the International Best Track Archive for Climate Stewardship (IBTrACS) were used to analyze the tracks of typhoons in the Western Pacific Basin, especially those that have passed through the Tropical Cyclone Advisory Domain (TCAD) and Philippine Area of Responsibility (PAR) [17]. The IBTrACS (v4r01) Western Pacific subset used in this study is available from NOAA NCEI [18]. The TCAD is the domain that encloses the PAR, which includes the area bounded by the imaginary lines connecting the coordinates 4°N 114°E, 27°N 114°E, 27°N 145°E, and 4°N 145°E but does not include the PAR with coordinates 5°N 115°E, 15°N 115°E, 21°N 120°E, 25°N 120°E, 25°N 135°E, and 5°N 135°E [19]. Typhoons from May 2000 up to November 2025 in the Western Pacific Basin were included, with 51,127 data points across 792 tropical cyclones.

Impact data for each typhoon were scraped from official NDRRMC situation reports for the same period. These included figures for affected population, evacuated families, infrastructure disruptions, casualties, and declarations of states of calamity. Local vulnerability indicators were sourced from two publicly available datasets: (1) 2018 city-level poverty incidence estimates from the Philippine Statistics Authority [15], and (2) model-based estimates of poor household counts derived using Poisson regression [16]. Geographic flood risk data were retrieved from Project NOAH and aggregated by city [20].

3.2 Storm-Scale Module

3.2.1 Track Clustering

For clustering of the tropical cyclone tracks near the Philippines, grid-based track representation combined with the K-means algorithm. Tropical cyclone tracks were first sorted by their storm identifier (SID) and by timestamp (ISO.TIME). The dataset was then filtered to only points within TCAD and PAR, defined as having a longitude between 114°E and 145°E and latitude between 4°N and 27°N. Furthermore, the presence of storm points in an imaginary 6 x 9 grid covering the Philippines, longitude between 116°E and 128°E with 6 divisions and latitude between 4°N and 22°N with 9 divisions, making 54 squares that represents a possible area traversed by a tropical cyclone. For each storm track, a binary presence vector with length 54 represents passing through a square at least once (encoded as “1”) and all other squares were set at “0”. This was done to create a fixed-length representation for all tropical cyclones regardless of the number of track points in the dataset.

The binary presence vectors for each typhoon were assembled into a feature matrix used for K-means clustering using scikit-learn. An arbitrary number of 10 clusters was chosen. Each cluster represents a distinct track of where the tropical cyclones usually pass through. Tropical cyclones that never entered the grid (vector with all “0”) were assigned in a separate Cluster 0, which assumes that these tropical cyclones were far enough to directly interact with the Philippines. Visualizations were produced for each cluster using Cartopy and Matplotlib, plotting all tracks belonging to each cluster.

3.2.2 Rapid Intensification Modeling

For identifying rapid intensification of tropical cyclones near the Philippines, data for each typhoon was segmented into multiple 24-hour intervals, each with 8 timestamps having 3-hour difference. Missing feature values were filled with 0 instead of dropping them. Tropical cyclones with less than 8 recorded data points were excluded. Numerical features for the timestamps, except for maximum sustained wind speeds to avoid data leakage, were used and flattened into a single feature vector for each interval. Rapid intensification labels were computed from the maximum sustained wind speed at the last timestamp and the next 24 hours. If the maximum sustained wind speed increases by at least 30 knots, then the interval is labeled as rapid intensification “1”, otherwise no rapid intensification “0”. The feature matrix and labels were shuffled and split into stratified training (60%), validation (20%), and test (20%) sets to preserve class proportions and then scaled. Synthetic Minority Oversampling Technique (SMOTE) was used to address the class imbalance on the train set. Naive Bayes (NB), Support Vector Machine (SVM), and Artificial Neural Network (ANN) classifiers were used to predict rapid intensification. Naive Bayes assumes each feature is normally distributed and a small number 1e-09 was used to avoid division by zero. SVM used the Radial Basis Function (RBF) as kernel. ANN used a multilayer perceptron with 2 hidden layers of sizes 32 and 16, respectively, with ReLU activation functions for all layers. Metrics used were Accuracy, Precision, and Recall.

3.2.3 Bantay-Bagyo Index (BBI)

Feature Engineering

Constructing the Classification Target

City-level typhoon impact reports from the NDRRMC were programmatically scraped, cleaned, and consolidated across all relevant storm events. Key variables included: percentage of barangays affected, proportion of affected families and population (normalized by city population), number of injured, dead, and missing persons per 100,000 residents, and binary indicators for class suspensions, work suspensions, flooding, power interruptions, and states of calamity declarations.

To construct the classification target, a composite severity score was created by assigning weights to three categories of impact: exposure indicators, service and mobility disruptions, and human cost indicators. Each component was processed and normalized before being aggregated into an overall impact severity score.

1) Exposure Indicators. These included affected population and evacuation rates, capturing the scale of community disruption. Each indicator was clipped to the [0, 100] range, then normalized to [0, 1]:

$$\text{Affected (\%)} = \frac{\text{Affected population}}{\text{City population}} \times 100 \quad (1)$$

$$\text{Evacuated (\%)} = \frac{\text{Evacuated population}}{\text{City population}} \times 100 \quad (2)$$

$$x_{\text{norm}} = \frac{x_{\text{clipped}} - \min(x)}{\max(x) - \min(x)} \quad (3)$$

The normalized exposure components were averaged to form the exposure index:

$$\text{Exposure Index} = \frac{1}{5} \sum_1^5 x_{\text{norm}} \times 100 \quad (4)$$

2) Service and Mobility Disruptions. Binary indicators such as class and work suspensions, power interruptions, flooding, and state of calamity declarations were averaged:

$$\text{Disruption}_{\text{binary}} = \frac{1}{5} \sum_1^5 \mathbf{1}(\text{disrupted}_j) \quad (6)$$

Casualty indicators were computed per 100,000 population:

$$x_{\text{rate}} = \frac{x}{\text{population}} \times 100,000 \quad (7)$$

These were normalized and combined with weights $w = [0.2, 0.6, 0.2]$ for injured, dead, and missing respectively:

$$\begin{aligned} \text{Casualty Component} = & 0.2 \cdot \text{Injured}_{\text{norm}} + 0.6 \cdot \text{Dead}_{\text{norm}} \\ & + 0.2 \cdot \text{Missing}_{\text{norm}} \end{aligned} \quad (8)$$

The overall disruption index was:

$$\begin{aligned} \text{Disruption Index} = & \frac{\text{Disruption}_{\text{binary}} + \text{Casualty Component}}{2} \\ & \times 100 \end{aligned} \quad (9)$$

Table 1: Summary of Typhoon Track Clustering, Rapid Intensification, and Human Impact Studies

Study	Method	Key Findings	Gap Addressed by BBI
Elsner & Liu, 2003, “Examining the ENSO–TC hypothesis” [5]	K-means clustering of WNP typhoon tracks based on peak and ending positions	Clusters align with ENSO phases but track shapes are simplified	Goes beyond static clustering by integrating full track shapes, intensity evolution, and impact modeling
Camargo et al., 2007, “Cluster analysis of typhoon tracks. Part I: General properties” [6]	Regression mixture models to cluster variable-length typhoon tracks	Linked track clusters to climate patterns like ENSO	Adds impact modeling by connecting track clusters with intensity and exposure to estimate local effects
Kim et al., 2011, “Pattern classification of typhoon tracks using fuzzy c-means clustering” [7]	Fuzzy c-means clustering to allow partial membership in multiple track types	Captured continuity between straight and recurving storms	Links track type to intensity trends and human impact, which this track-only study does not
Yan et al., 2025, “Assessment and application of tropical cyclone clustering in the South China Sea” [8]	Compared K-means, fuzzy c-means, and SOM for classifying tracks	Identified track types linked to ENSO and rainfall patterns	Translates track types into localized risk via exposure and vulnerability—this study stops at hazard patterns
Mesias & Bagtasa, 2025, “AI-Based Tropical Cyclone Rainfall Forecasting in the Philippines Using Machine Learning” [20]	SOM clustering of TC tracks + Random Forest regression for rainfall prediction	AI model using TC track vectors, translational speed, duration produced rainfall predictions comparable to satellite data	Integrates rainfall prediction with track clustering and human impact, extending hazard-only studies
Chang et al., 2020, “Self-organizing maps of typhoon tracks allow for flood forecasts up to two days in advance” [21]	Grid-based TC track vectors + SOM; linked to hydrology for flood forecast	Track clustering corresponded to runoff/flood response; enables 2-day ahead flood forecasts	Adds localized hydrologic impact prediction using track archetypes
Fang et al., 2022, “A Novel Tropical Cyclone Track Forecast Model Based on Attention Mechanism” [22]	ConvGRU deep-learning with spatial grids for TC center location	Grid-based representation improved track forecast performance	Improves TC track forecast skill to support downstream impact estimation
Xu et al., 2025, “Classification of Tropical Cyclone Tracks in the Northwest Pacific Based on the SD-K-Means Model” [23]	Soft-DTW + K-Means clustering, including motion features	Identified 4 clusters capturing both spatial and dynamic motion behaviors	Captures dynamic track behavior for improved risk modeling
Kaplan et al., 2010, “A revised tropical cyclone rapid intensification index” [9]	NOAA’s operational RI Index using SHIPS predictors	Provided probabilistic forecasts of RI within 24 hours	Uses RI forecasts to project human impact—something RI Index alone does not do
Rozoff & Kossin, 2011, “Probabilistic forecast models for tropical cyclone rapid intensification” [10]	Machine learning (Naive Bayes, logistic regression) to improve RI forecasting	Improved detection of RI using non-linear probabilistic models	Incorporates RI probabilities into impact estimation, not just hazard prediction
Kim et al., 2024, “Predicting rapid intensification of tropical cyclones in the western North Pacific” [24]	Ensemble ML models with Net Energy Gain Rate (NGR) as predictor	Ensemble models improved RI prediction accuracy	Translates forecasted RI events into predicted LGU-level impact using exposure and vulnerability
Qu et al., 2025, “Accurate tropical cyclone intensity forecasts using a non-iterative spatiotemporal transformer model” [11]	Transformer-based model (TIFNet) to forecast intensity trajectories	Outperformed existing models, especially for RI	Uses improved intensity forecasts as inputs to impact models—extending their value beyond hazard level
Peduzzi et al., 2012, “Global trends in tropical cyclone risk” [12]	Disaster risk modeled as hazard $\times$ exposure $\times$ vulnerability	Governance and development strongly influence mortality	Applies this framework to real-time impact forecasting, not just long-term trend analysis
Tennant & Gilmore, 2020, “Government effectiveness as a determinant of cyclone mortality” [13]	Statistical analysis of cyclone mortality vs governance indicators	Lower mortality in regions with stronger governance, independent of storm intensity	Operationalizes governance insights by factoring local vulnerability and preparedness into predictive models
Vessey et al., 2025, “Predicting North Atlantic hurricane damage” [14]	Hazard $\times$ exposure $\times$ vulnerability model, preprint	Reduced prediction error vs. wind-speed-only models	Adapts approach to the Philippines, includes multiple impact types, integrates dynamic hazard inputs like RI and track clusters

Table 2: Summary of features used in modeling city–storm impact severity.

Feature	Description
city_std	Standardized name of the Metro Manila city
year	Calendar year of the typhoon event (e.g., 2020, 2021)
typhoon	Local/common name of the typhoon associated with the impact data
combined_effect	Boolean flag indicating whether the reported impacts are due to a combined event (e.g., typhoon + southwest monsoon, or multiple storms) rather than a single isolated typhoon
combined_with	Text description of the secondary system in a combined event (e.g., “SW Monsoon”, another typhoon name); NaN if <code>combined_effect</code> is False
SID	Storm identifier from IBTrACS (or equivalent best-track dataset), uniquely identifying the tropical cyclone
NAME	International / basin-standard storm name from IBTrACS (may differ from local PAGASA name)
WIND_max	Maximum sustained wind speed of the storm (knots) over its lifetime, taken from track data
WIND_ave	Average sustained wind speed (knots) over the storm’s lifetime (mean of track points)
PRES_min	Minimum central pressure (hPa) recorded for the storm
R34_max	Maximum radius of 34-knot winds (nautical miles) across quadrants – proxy for storm size at gale-force threshold
R50_max	Maximum radius of 50-knot winds (nautical miles) across quadrants – proxy for storm size at storm-force threshold
R64_max	Maximum radius of 64-knot winds (nautical miles) across quadrants – proxy for storm size at typhoon-force threshold
POCI_ave	Average Pressure of the Outermost Closed Isobar (POCI, hPa) over the track – measure of outer pressure field / storm environment
ROCI_ave	Average Radius of the Outermost Closed Isobar (ROCI, nautical miles) – approximate outer size of the storm system
RMW_ave	Average Radius of Maximum Wind (RMW, nautical miles) over the track – typical distance from center to strongest winds
GUST_max	Maximum gust wind speed (knots) recorded for the storm
STORM_SPEED	Average translational / forward speed of the storm (usually knots or km/h) over the track
STORM_DIR	Average direction of storm motion in degrees (0–360, meteorological azimuth)
storm_cluster	Numeric label indicating the storm archetype cluster (e.g., from unsupervised clustering on IBTrACS features)
NATURE	Storm classification string (e.g., “TROPICAL DEPRESSION”, “TROPICAL STORM”, “TYPHOON”)
has_RI	Indicator (“True”/“False”) for whether the storm experienced at least one Rapid Intensification (RI) episode over its lifetime
first_RI_window_index	Index of the first time window in the track where RI occurred (e.g., 24-h $\Delta V \geq$ threshold), relative to the ordered best-track points; NaN if no RI
city_pct_area_5yr_low	Percentage of the city’s land area within low flood hazard zones for a 5-year return period (from NOAA or similar)
city_pct_area_5yr_med	Percentage of the city’s land area within medium flood hazard zones for a 5-year return period
city_pct_area_5yr_high	Percentage of the city’s land area within high/severe flood hazard zones for a 5-year return period
poor_households_2018	Number of poor households in the city (around 2018 baseline; from official poverty statistics)
poverty_rate_2018	Poverty incidence in 2018 (percentage of households/individuals classified as poor) for the city
impact_severity_3band	Target label: 3-band impact severity class for the city–storm pair (0=Low, 1=Medium, 2=High) constructed from exposure, disruption, and human-impact indicators
season	Climatological season of storm occurrence in the Philippines, derived from UTC event time. Season is categorized as DJF (Dec–Feb, NE monsoon), MAM (Mar–May, dry/hot season), JJA (Jun–Aug, SW monsoon peak), or SON (Sep–Nov, late monsoon with frequent strong typhoons)

3) Composite Impact Severity. The final impact score was a weighted average of exposure and disruption severity:

$$\text{Impact Severity} = 0.6 \cdot \text{Exposure Index} + 0.4 \cdot \text{Disruption Index} \times 100 \quad (10)$$

These continuous scores were then classified into Low, Medium, or High impact bands using the 25th and 75th percentiles of the training data (2020–2024) as bin thresholds to avoid data leakage during evaluation.

The final impact severity label was defined as a three-class ordinal variable: low, medium, or high. Percentile thresholds for these categories were computed using only the training set (i.e., typhoon events from 2020 to 2024), in order to prevent data leakage. These thresholds were then used to bin the test set (2025 storms), ensuring that model evaluation was based on unseen data labeled with consistent, non-circular criteria. This labeled target variable served as the ground truth for supervised learning, enabling the Bantay-Bagyo Index (BBI) model to predict impact severity based on pre-landfall storm features, geographic exposure, and vulnerability metrics.

City-Level Flood-Hazard: 5 year return Flood hazard data were processed from Project NOAH’s raster maps of flood susceptibility. The 5-year return period hazard layer was selected as a moderate-risk scenario representative of typical storm-related flooding, balancing severity and recurrence. Each city’s shapefile boundary was overlaid with the hazard map, and the percentage of area falling within low, medium, and high flood zones was calculated. These three proportions were used as continuous features representing geographic exposure.

Storm Features Storm-specific features included maximum sustained wind, minimum central pressure, average radius of 34-, 50-, and 64-knot wind fields, average radius of maximum winds, pressure outer core index, and storm translational speed and direction. Each storm was also assigned a cluster label based on its track pattern and tagged with rapid intensification (RI) status and number of RI episodes.

Social Vulnerability Features Vulnerability features included city-level poverty rate (as a percentage) and estimated poor household counts from the Poisson regression model. Hazard exposure was represented by the three flood zone area proportions. All features were merged into a unified event-level dataset indexed by storm and city.

Modeling the Bantay-Bagyo Index (BBI)

The final dataset combined storm characteristics, flood exposure, and socioeconomic vulnerability features for each city–storm pair, paired with the corresponding three-class impact severity label. Table 2 summarizes all the features used in modeling. All preprocessing — including encoding categorical variables and scaling numerical ones — was integrated into a `scikit-learn` [29] pipeline to streamline model training and evaluation.

To model impact severity, several classifiers were explored, including Gaussian Naive Bayes, Support Vector Machines

(SVM), Perceptron, AdaBoost, and XGBoost. Models were trained using typhoons from 2020–2024 and evaluated on 2025 events to ensure temporal generalization. Stratified cross-validation was used to preserve class balance during training, and performance was assessed using accuracy, precision, recall, and F1-score. To address the class imbalance, and to make sure the model never “underprepares” people for future typhoons, the model is tuned to find the right class weights that prioritizes the recall for class 2 (“high” severity) label. Feature importance analysis was performed to identify which storm and vulnerability indicators most influenced predicted impact levels, offering insight into the drivers of local disaster severity under typhoon conditions.

4. RESULTS AND DISCUSSION

4.1 Tropical Cyclone Track Clustering

The tropical cyclone track clusters show that the typhoons do not behave randomly but follow distinct pathways. Cluster 1 typhoons originate from the south-eastern side of the Philippines traveling to north-west, approaching Luzon but mostly stay near the coastline. Cluster 2 typhoons do not group well with the other clusters, but are close enough to affect the Philippines. Cluster 3 typhoons originate from the south-eastern side of the Philippines, hit Leyte and travel west-northwest through other provinces in Visayas. Cluster 4 typhoons approach at a similar angle to Cluster 3, but at higher latitudes hitting Samar, Bicol, and Southern Tagalog. Cluster 5 typhoons also follow a similar angle to Clusters 3 and 4, but at much higher latitudes grazing the northern tip of Luzon and hitting Batanes. Cluster 6 typhoons originate from the east-southeast and travel west-northwest, hitting the northeastern side of Luzon, the Cagayan Valley. The Sierra Madre Mountains run along this eastern side, which is said to be an effective barrier for the Cagayan Valley as it is able to reduce both wind and rain exposure. However, extreme tropical cyclones can still affect the Cagayan Valley [28]. Cluster 7 typhoons originate from the eastern side of the Philippines, travel west-northwest hitting Batanes. Cluster 8 typhoons originate from the eastern side of the Philippines and travel west, passing through Central Luzon and Metro Manila. Cluster 9 typhoons originate from the eastern side and travel northwest, heading to Taiwan, but are still close enough to affect the Philippines. Cluster 10 typhoons originate from the southwestern side, traveling west and west-northwest, hitting Davao, Northern Mindanao, and Central Visayas before passing through Palawan and exiting the Philippines. The overall track patterns are visualized in Figure 1.

4.2 Rapid Intensification Prediction

Rapid Intensification (RI) prediction dataset derived from 24-hour intervals from IBTrACS consists of 168 features per interval with a total of 45,591 intervals. The dataset was split into 27,354 train, 9,118 validate, and 9,119 test intervals. SMOTE was applied to the training set to balance the classes, which initially had 5,295 labeled as “1” and 22,059 labeled as “0”. After SMOTE, both classes have 22,059 intervals.

For Naive Bayes, both the Accuracy and Precision are low but the Recall is high across the train, validate, and test sets. This indicates that the model labels a lot of events as RI,

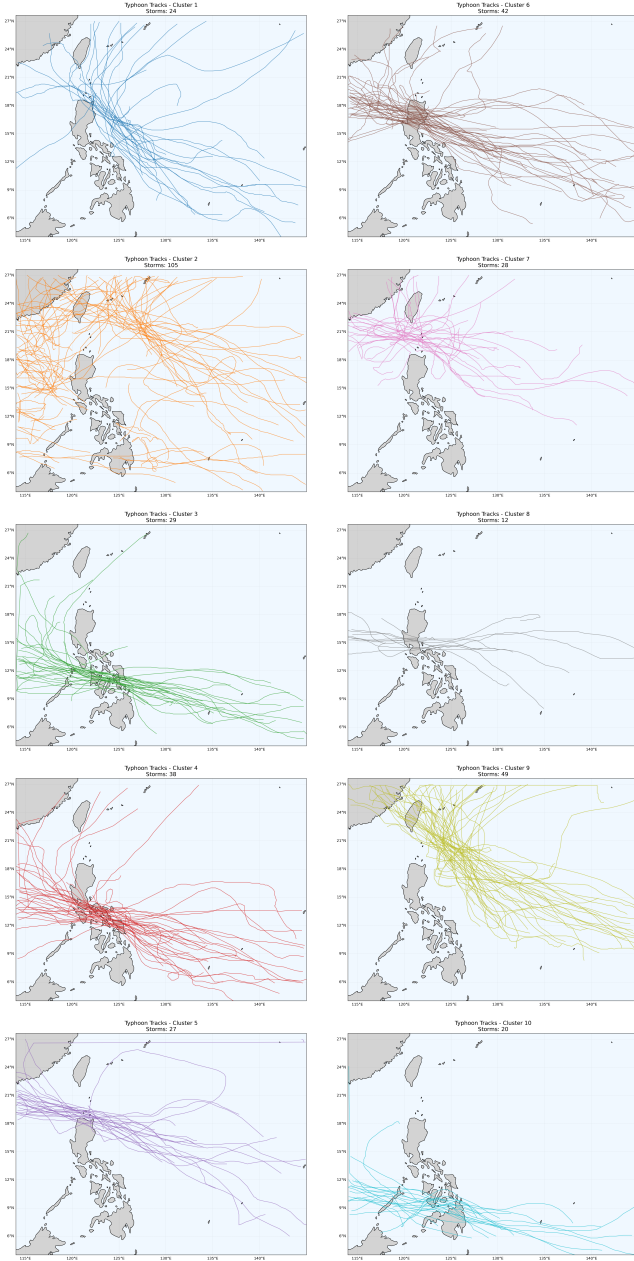


Figure 1: Tropical Cyclone Tracks per Cluster in the Philippines from 2000-2025.

correctly identifying most of them but producing many false positives. For SVM, the Precision is low but Accuracy and Recall are high, showing an improvement over Naive Bayes while still generating some false positives. ANN shows the best overall performance among the three classifiers. Table 3 summarizes the metrics for each classifier and set, while Table 4 shows the confusion matrices on the test set for each classifier.

To develop the BBI model, the impact severity target was first standardized into a three-band categorical variable (Low, Medium, High). Figure 2 presents the continuous impact severity scores (0–100) for each typhoon–city pair in Metro

Table 3: Accuracy, Precision, and Recall metrics for each classifier on Train, Validate, and Test sets

Classifier	Set	Accuracy	Precision	Recall
Naive Bayes	Train	0.592	0.308	0.890
	Validate	0.595	0.309	0.884
	Test	0.595	0.310	0.894
SVM	Train	0.863	0.592	0.941
	Validate	0.852	0.574	0.910
	Test	0.848	0.567	0.898
ANN	Train	0.952	0.839	0.933
	Validate	0.895	0.703	0.791
	Test	0.898	0.703	0.816

Table 4: Confusion matrices on the Test set for each classifier

Naive Bayes	True \ Pred	1	0
	1	1578	187
	0	3506	3848
SVM	True \ Pred	1	0
	1	1585	180
	0	1208	6146
ANN	True \ Pred	1	0
	1	1441	324
	0	609	6745

Manila. The heatmap highlights Typhoon Carina as the most damaging recent event: even its least affected city, Quezon City, recorded a score of 12.4, while Marikina City reached 71.0, the highest single-city impact in the dataset. Across all storms in our dataset, Marikina consistently appears in the upper range of the impact severity scale, a result consistent with previous hydrologic and disaster literature identifying the city as one of the most flood-prone local government units in Metro Manila. Studies of the Marikina River Basin show frequent, high-magnitude river floods affecting residential and commercial zones [30, 31], while official disaster reports describe Marikina as among the most severely impacted during major events such as Typhoon Ulysses (2020) and Tropical Storm Ondoy (2009) [32, 33].

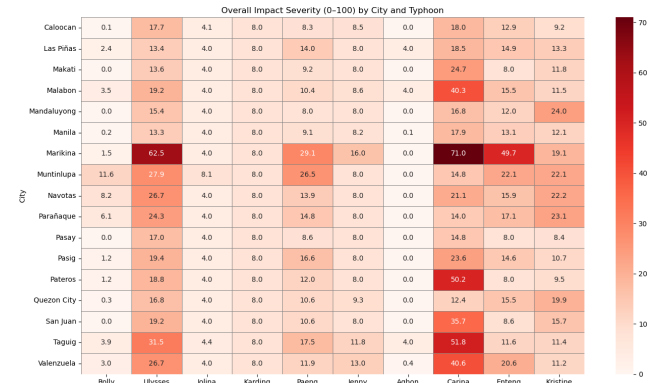


Figure 2: Impact severity heatmap by City in Metro Manila and Typhoon.

Figure 3 shows the same city–typhoon matrix but discretized into the three impact bands used as labels for model training. Carina again stands out, with High severity recorded in

multiple eastern and low-lying cities (e.g., Marikina, Pasig, Taguig, Valenzuela). Several other storms, such as Paeng, Ulysses, and Rolly, also generate widespread Medium to High severity across the metropolis, while weaker systems (e.g., Aghon) are mostly confined to Low impacts.

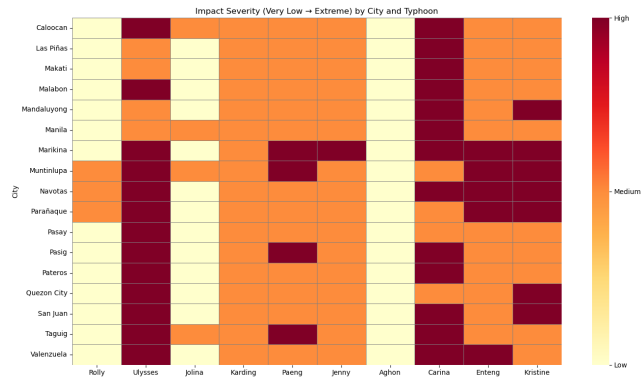


Figure 3: Discretized city-typhoon impact matrix showing three impact bands (Low, Medium, High) used as labels for model training.

The seasonal profile of these events is summarised in Figure [X] and Table [X]. Table [X] lists the storms in the modelling dataset by season:

The seasonal profile of these events is summarized in Figure 4 and Table 5. Table 5 lists the storms in the modelling dataset by season:

Table 5: Storms in the modelling dataset by season.

Season	Typhoon(s)
JJA	Carina
MAM	Aghon
SON	Enteng, Jenny, Jolina, Karding, Kiko, Kristine, Paeng, Rolly, Ulysses, Uwan

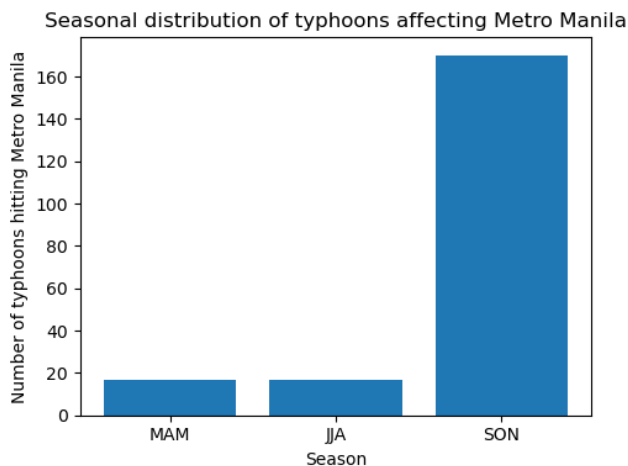


Figure 4: Seasonal profile of typhoon impacts in the dataset.

Of the 12 typhoons that significantly affected Metro Manila and are included in the BBI dataset, 10 occurred in SON

(September–November), while only one occurred in MAM (Aghon) and one in JJA (Carina). This pattern is consistent with the regional climatology, where late-monsoon storms during SON tend to be stronger and more impactful over Luzon.

Finally, the overall distribution of the three-band target is:

- Medium: 95 observations (46.6%)
- Low: 61 observations (29.9%)
- High: 48 observations (23.5%)

This indicates a moderately imbalanced but still informative target: Medium severity is the most common outcome, while both Low and High classes are sufficiently represented to support supervised learning, especially when combined with class-sensitive training (sample weighting) and evaluation that explicitly prioritizes performance on the High-severity class.

To assess predictive relationships and prevent redundancy among inputs, a correlation matrix was computed using the merged dataset of storm features, city-level vulnerability indicators, and hazard exposure variables. Each row represented a unique typhoon–city event, labeled with its corresponding impact severity. The matrix revealed clusters of highly correlated variables, particularly among storm structure indicators (e.g., wind radii and maximum wind speed).

To mitigate multicollinearity and streamline the modeling process, features with high pairwise correlations were reviewed and selectively removed. The resulting feature set, visualized in the correlation heatmap, represents a refined, non-redundant group of predictors that preserves explanatory diversity without compromising statistical validity.

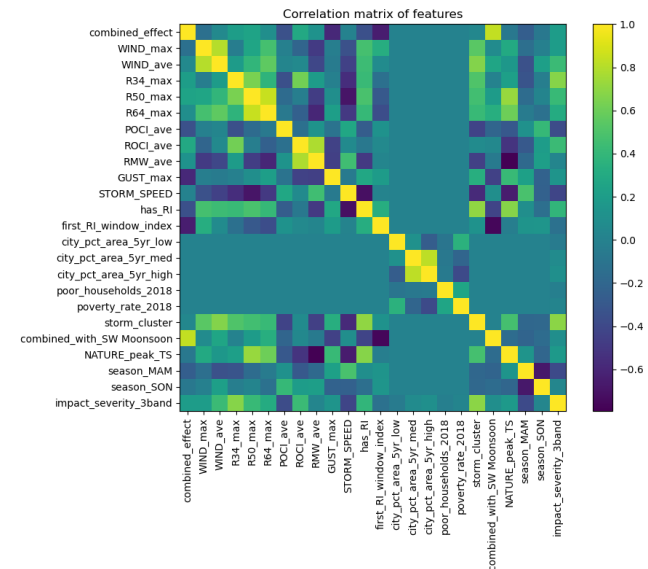


Figure 5: Correlation matrix of features.

The dataset was divided into a training set (typhoons from

2020 to 2024) and a test set (typhoons from 2025) to avoid temporal data leakage during evaluation. Class distribution in the training set was relatively balanced across impact severity levels: 26% low, 49% medium, and 25% high. This stratification was reflected in year-wise class distribution and used to compute class weights for algorithms sensitive to imbalance, such as SVM and Perceptron. After feature standardization, several classifiers were trained: Naive Bayes, SVM, Perceptron, AdaBoost with Perceptron, and XGBoost.

Performance varied significantly across models. XGBoost achieved the best accuracy (85.3%) and macro F1-score (0.710), with high precision and recall for low- and medium-impact classes but lower recall on the high-impact class. In contrast, the Perceptron and AdaBoost+Perceptron models exhibited poor recall for medium-impact events, though the Perceptron correctly classified all high-impact events. SVM and Naive Bayes models favored the medium class, misclassifying most low-impact events. These results, shown in the confusion matrices, indicate that tree-based models like XGBoost are more effective in capturing the complex patterns between pre-landfall typhoon characteristics and localized impact severity.

Although the XGBoost model achieved the highest overall accuracy (85.3%) and macro F1-score (0.710), it underperformed in identifying high-severity (class 2) events — correctly predicting only 1 out of 5 instances (recall = 0.200). Given that preparedness for extreme impacts is critical, especially to avoid under-response, the study prioritized improving recall for class 2. To address this, class imbalance was handled by adjusting class weights using a tuning parameter λ , where class 2 was assigned a weight of λ and class 1 was given $\lambda/2$, keeping class 0 fixed at a weight of 1.

A series of models were trained using λ values from 1 to 10. Performance stabilized at $\lambda \geq 3$, with a consistent improvement in class 2 recall from 0.200 to 0.800 and F1-score rising from 0.333 to 0.667. Precision for class 2 also remained stable at approximately 0.571, and macro F1 peaked at 0.789. These models correctly identified 4 out of 5 class 2 events while maintaining balanced classification for lower-impact cases. Based on this tradeoff between improved high-severity recall and overfitting risk, the final model adopts $\lambda = 4$ (weights: class 0 = 1.0, class 1 = 2.0, class 2 = 4.0).

The calibrated XGBoost model, using $\lambda = 4$ for class weighting (weights: Class 0 = 1, Class 1 = 2, Class 2 = 4), achieved a strong balance between overall accuracy and recall on high-severity cases. It attained an overall accuracy of 85.3% and a macro-averaged F1-score of 0.822, significantly improving upon prior models in both general performance and critical class sensitivity.

As shown in the confusion matrix, the model correctly identified 4 out of 5 high-severity (Class 2) events, yielding a recall of 0.80 and a precision of 0.667 for this class. This demonstrates the model’s improved capacity to detect potentially catastrophic storm impacts — a key objective in operational forecasting. Class 1 (moderate severity) also maintained a healthy precision of 0.769 and recall of 0.833, showing that the reweighting strategy did not overfit to the

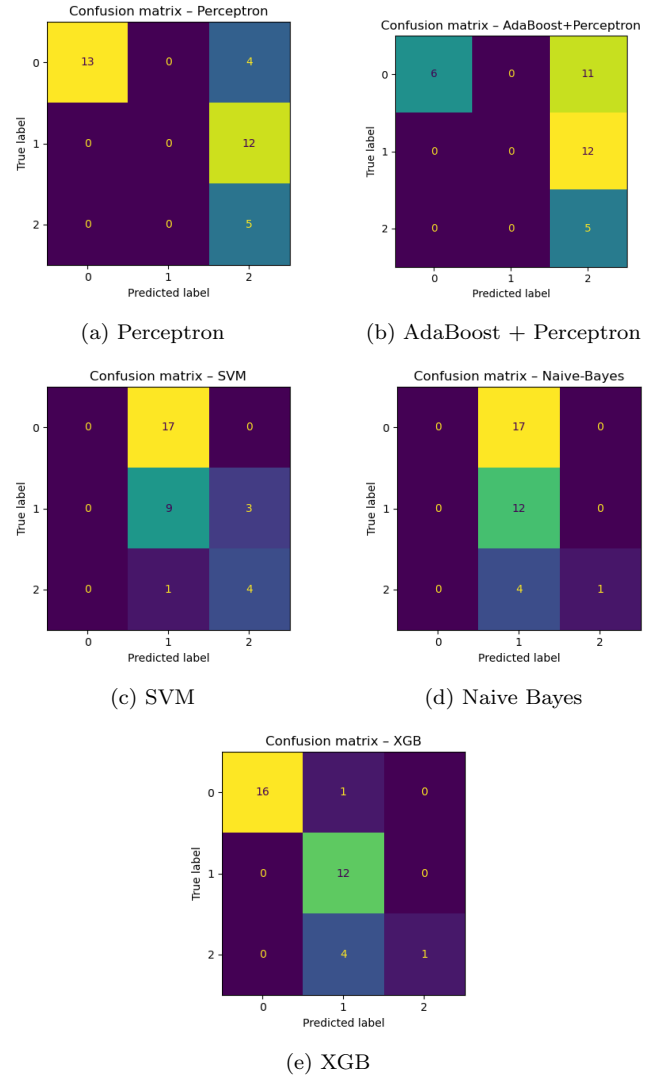


Figure 6: Confusion matrices for different classifiers. Each subfigure shows the predicted vs. true labels on the test set.

high-severity class. Low-severity cases (Class 0) were predicted with perfect precision and high recall (0.882), further reinforcing the model’s robustness across all classes.

This calibration thus offers a reliable operational balance: it avoids underwarning in dangerous situations, while retaining general predictive accuracy — making it suitable as the default configuration of the Bantay-Bagyo Index for impact-based early warning.

5. CONCLUSIONS AND FUTURE WORK

This study used machine learning to translate storm-scale information into LGU-level impact estimates, addressing a critical gap in current typhoon forecasting. Operational forecasts typically provide track, intensity, and rainfall warnings, but these are not directly expressed as expected local consequences that can trigger concrete actions such as evacuations, class and work suspensions, or pre-positioning of relief. By combining typhoon track clustering, rapid intensification (RI) modeling, and the *Bantay-Bagyo Index* (BBI),

Table 6: Performance metrics for classifiers on the 2025 test set by impact severity class. Classes: 0=Low, 1=Medium, 2=High.

Model	Recall (lr)4-6 (lr)7-9	Accuracy	Macro F1	Precision				
				0	1	2	0	1
XGBoost	0.200	0.853	0.710	1.000	0.706	1.000	0.941	1.000
Naive Bayes	0.200	0.382	0.289	0.000	0.364	1.000	0.000	1.000
SVM	0.800	0.382	0.376	0.000	0.333	0.571	0.000	0.750
AdaBoost + Perceptron	1.000	0.324	0.275	1.000	0.000	0.179	0.353	0.000
Perceptron	1.000	0.529	0.417	1.000	0.000	0.238	0.765	0.000

Table 7: Performance of XGBoost for varying class 2 weight λ

λ (Class 2)	Class 1 Weight	Precision (C2)	Recall (C2)	F1-Score (C2)	Macro F1
1.0	0.5	1.000	0.200	0.333	0.710
1.5	0.8	0.500	0.200	0.286	0.700
2.0	1.0	0.500	0.200	0.286	0.680
3.0	1.5	0.571	0.800	0.667	0.790
4.0	2.0	0.571	0.800	0.667	0.790
5.0	2.5	0.571	0.800	0.667	0.790
6.0	3.0	0.571	0.800	0.667	0.790
7.0	3.5	0.571	0.800	0.667	0.790
8.0	4.0	0.571	0.800	0.667	0.790
9.0	4.5	0.571	0.800	0.667	0.790
10.0	5.0	0.571	0.800	0.667	0.790

we moved toward an impact-based early warning framework for Metro Manila that is conceptually scalable to other regions of the Philippines.

K-means clustering was applied to spatial IBTrACS data, excluding other environmental factors, for a grid-based identification of dominant tropical cyclone pathways affecting the Philippines. The clustering produced interpretable typhoon track clusters that capture the paths originating from the southeastern side of the Philippines traveling northwest or west, crossing different provinces mostly of Visayas and Luzon. The clustering demonstrates that past typhoons follow a limited number of recurrent trajectories which can be used to infer the regions that will likely be exposed to future tropical cyclones. The method provides a simple data-driven approach to identifying tropical cyclone tracks that can be used for preparedness planning.

A machine learning framework was used to predict 24-hour rapid intensification of tropical cyclones using IBTrACS data. Initially, Naive Bayes, Support Vector Machine (SVM), and Artificial Neural Network (ANN) classifiers were trained and evaluated solely for the RI modeling before using more classifiers for the BBI model. Among these, ANN with two hidden layers showed the best accuracy of 86.9%, precision of 0.729, and recall of 0.829 for predicting rapid intensification.

We developed a three-class LGU impact model that links human disaster outcomes to storm characteristics, social vulnerability indicators, and 5-year flood hazard exposure. The final BBI classifier is a calibrated XGBoost model that

places the highest weight on high-impact events (Class 2) and half that weight on medium-impact events (Class 1). On the 2025 test storms, it achieved an overall accuracy of 85.3%, a macro-averaged F1-score of 0.822, and a recall of 0.80 for the high-impact class. This weighting strategy explicitly prioritizes correctly identifying severe events — consistent with preparedness goals where under-warning is more costly than over-warning — while maintaining strong performance on lower-impact classes.

Practically, the calibrated BBI model can be used to screen incoming storms, prioritize high-risk cities, and support early decisions on suspensions, evacuations, and resource deployment. Beyond response operations, the outputs can inform insurance pricing, disaster risk financing, and urban development by highlighting locations where structural mitigation and preparedness investments are most needed. Track cluster and RI information can also be embedded into impact-based warning messages (e.g., “this storm behaves like Cluster A and is RI-likely, expect higher BBI scores in Marikina”), and the framework lends itself to implementation in a live dashboard or decision-support tool for LGUs.

This work has several limitations. First, the dataset covers only 2020–2025 storms and focuses on Metro Manila, so the number of training events is limited and geographic generalization has not yet been tested. Second, impact reporting from NDRRMC may be incomplete or inconsistent across cities, introducing noise into the ground-truth labels. Third, the hazard representation is predominantly wind-centric, with flood risk approximated by a single 5-

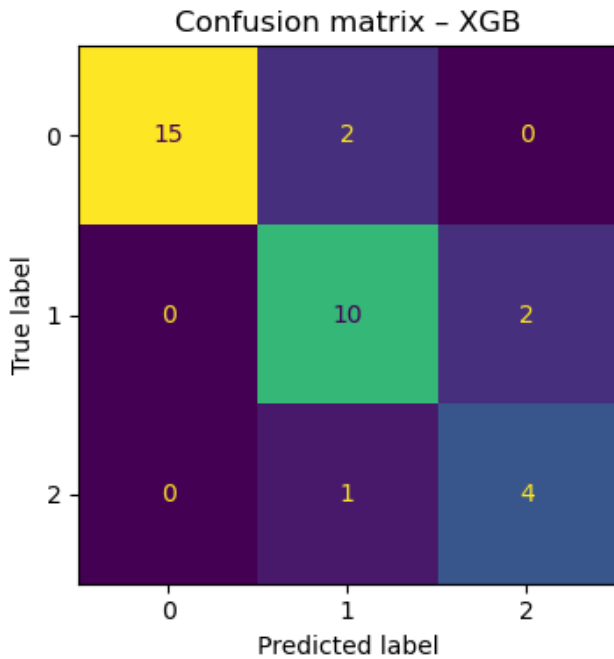


Figure 7: Confusion matrices for calibrated XGBoost model

year return-period layer from Project NOAH, rather than explicit rainfall-runoff or storm-surge modeling. Finally, the model is calibrated on Metro Manila conditions, and its performance for other regions with different exposure and vulnerability profiles remains to be validated.

Future work should extend BBI to other regions and LGUs nationwide, validating and recalibrating the model under diverse hazard and vulnerability contexts. Incorporating additional environmental predictors — such as sea surface temperature, vertical wind shear, and gridded rainfall fields — could improve both RI and impact modeling. Beyond generic track clusters, it would be valuable to define archetypal “storm types” (e.g., Ondoy-like rainfall extremes versus wind-dominant events) to tailor warnings for specific hazards and housing conditions. Finally, coupling the framework with real-time PAGASA forecast tracks and enriching the vulnerability layer with information on building type, informal settlements, and preparedness indicators would move the system closer to an operational impact-based early warning tool for the Philippines.

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